

Spring 2026 Topic Reviews for Math 54

Faculty: Profs. Arun Sharma and Vera Serganova
Facilitator: Jason Ferguson, jason.ferguson@berkeley.edu
Time and Location: Tu 3:30-5 on Zoom and in 201A Student Center, Th 3:30-5 on Zoom and 6 Evans
Jason's Office Hours: As of 3/16, M 12-1, TuTh 2-3 [See here](#) for the most up-to-date Drop-In Schedule



Practice Final for 3:30-5pm Thursday, May 7, 2026's Math 54 Problem-Solving Session

Objectives: By engaging actively with this session, participants will

1. Assess in real-time their problem-solving skills when solving individual novel, Final Exam Exam-level problems in an exam setting
2. Identify types of problems that would benefit them to practice more

Note: I have not seen Prof. Serganova's OR Prof. Sharma's Math 54 Final yet. I tried to make this Practice Final be representative of what I believe to be the length, question types, and difficulty of the their Finals, but cannot guarantee that that is the case. In particular, I cannot guarantee this Final is comprehensive. In other words the actual Final may ask different types of questions and may cover content not on this Practice Final.

Notice: Prof. Sharma's final has 13 problems (9 free-response, 3 sets of 5 true/false with no explanation, and 1 bonus), while Prof. Serganova's final has 9 problems (2 of which are true/false).

This practice final has 12 problems (8 free-response, 3 sets of 5 true/false, and 1 bonus/harder question), so it does not exactly match the question number or type of problem in either final, but the overall length feels representative.

Also notice: There are two problem 7s and two problem 8s; one intended for Prof. Serganova's students and one intended for Prof. Sharma's students.

Breakdown:

- Free-response: Problems 1-6 (25 pts each)
- Free-response, but different problems for different lectures Problems 7AB-8AB (25 pts each)
- True/False: Problems 9-11 (25 points each)
- Free-response/bonus: Problem 12 (15 points each)

Print your name: _____

SID Number: _____

TIME AND CONDITIONS: 170 minutes; (for Prof. Sharma) one note and formula sheet provided no other notes allowed (for Prof. Serganova) bring a formula sheet you create; closed book/internet; no calculator/computer

Space for notes on test-taking strategies:

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1. (25pts) Let A be the following matrix:

$$\begin{bmatrix} 0 & 1 & 3 & 0 \\ 1 & 0 & 0 & 4 \\ 0 & 0 & 0 & 2 \\ 0 & 2 & 5 & 2 \end{bmatrix}$$

Do the following parts in any order you wish.

- (a) Compute $\det A$
- (b) Compute A^{-1} .

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2. (25pts) Let V be the vector space of all twice-differentiable functions on $(-\infty, \infty)$. Let W be the vector space of all functions on $(-\infty, \infty)$. Let $T: V \rightarrow W$ be the function:

$$T(y) = y'' + 4y' + 5y$$

- (a) Find a basis for the kernel of T .
- (b) Find all solutions y to the following equation using undetermined coefficients:

$$T(y) = e^{-2t} + \sin(t)$$

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3. (25pts total) Consider the following system of linear equations, where a is a constant:

$$\begin{aligned}x_1 + x_2 - x_3 &= a \\x_1 + ax_2 + 2x_3 &= a \\2x_1 + 2x_2 + ax_3 &= -4\end{aligned}$$

Do the following in any order:

- (a) Find all numbers a for which the system is consistent.
- (b) Find all values of a for which the system has infinitely many solutions.
- (c) Choose one of the values a for which the system has infinitely many solutions (there is at least one). For that value a , write the solution to the system in parametric vector form.

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4. (25pts) Find a singular value decomposition of the following matrix A :

$$A = \begin{bmatrix} -2 & 4 \\ 2 & -4 \\ 1 & -2 \end{bmatrix}$$

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5. (25pts) Let:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix},$$

and $W = \text{Col } A$. Let $T: \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be the orthogonal projection onto W .

- (a) Find the matrix of T . (For partial credit, find $T(\mathbf{y})$.)
- (b) Explain why $A^T(\mathbf{p} - T(\mathbf{p})) = \mathbf{0}$ for any $\mathbf{p} \in \mathbb{R}^4$. (The “ T ” in “ A^T ” is “transpose”.)

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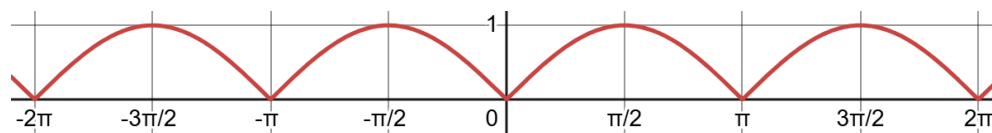


6. (25pts) Let $f(x) = \sin(x)$.

- (a) Find the Fourier sine series of $f(x)$ on $[0, \pi]$.
- (b) Show using Fourier series that for all x ,

$$\frac{2}{\pi} - \frac{4}{\pi} \sum_{\substack{k \text{ even,} \\ k \geq 2}} \frac{1}{k^2 - 1} \cos(kx) = |\sin(x)|.$$

The graph of $|\sin(x)|$ is shown below:



The identities $\sin(a + b) = \sin(a) \cos(b) + \cos(a) \sin(b)$ and $\sin(a - b) = \sin(a) \cos(b) - \cos(a) \sin(b)$ may be helpful.

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7A (25pts) **For Prof. Serganova's lecture only; Prof. Sharma's students should solve 7B on the next page.** [This question is fair for Prof. Sharma.]

Let $\mathbf{a}$ and $\mathbf{b}$ be any vectors in $\mathbb{R}^n$ with $\mathbf{a} \cdot \mathbf{b} \neq 0$. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the function

$$T(\mathbf{x}) = \frac{\mathbf{a} \cdot \mathbf{x}}{\mathbf{a} \cdot \mathbf{b}} \mathbf{b}$$

- (a) Show that T is a linear transformation.
- (b) Describe which vectors are in $\text{Ker } T$.
- (c) Find a basis for $\text{Ran } T$.
- (d) Let A be the matrix of T . Explain why A is diagonalizable, and find a matrix D for which there is an invertible matrix P with $A = PDP^{-1}$.

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7B (25pts) **For Prof. Sharma's lecture only; Prof. Serganova's students should solve 7A on the previous page.** [This question is fair for Prof. Serganova.]

Let $V = M_{2 \times 2}$, the vector space of all 2×2 matrices with real coefficients. Let H be the set of all symmetric 2×2 matrices.

- (a) Show that H is a subspace of V .
- (b) Find a basis for H and the dimension of H .

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8A (25pts) **For Prof. Serganova's lecture only; Prof. Sharma's students should solve 7B on the next page.** [Parts (a) and (b) can be solved using content Prof. Sharma covered, but not part (c).]

- (a) Let $\mathbf{A}(t)$ be any $n \times n$ matrix whose entries are functions that are continuous for $a < t < b$. Let $\mathbf{X}(t)$ be a fundamental matrix for the system of ODEs:

$$\mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t) \quad a < t < b.$$

Show that $\mathbf{X}'(t) = \mathbf{A}(t)\mathbf{X}(t)$.

- (b) Using part (a), find a 2×2 matrix $\mathbf{A}(t)$ of functions that are continuous on $(0, 2)$ for which $\begin{bmatrix} e^{5t} \\ 2e^{5t} \end{bmatrix}$ and $\begin{bmatrix} t \\ t^2 \end{bmatrix}$ are both solutions to $\mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t)$.
- (c) Explain why there is no 2×2 matrix $\mathbf{A}(t)$ of functions that are continuous on $(-\infty, \infty)$ for which $\begin{bmatrix} e^{5t} \\ 2e^{5t} \end{bmatrix}$ and $\begin{bmatrix} t \\ t^2 \end{bmatrix}$ are both solutions to $\mathbf{x}'(t) = \mathbf{A}(t)\mathbf{x}(t)$.

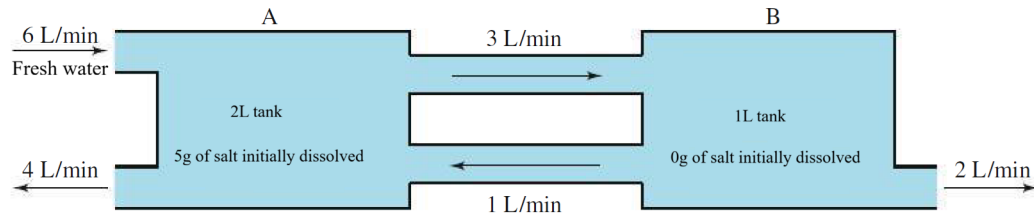
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8B (25pts) **For Prof. Sharma's lecture only; Prof. Serganova's students should solve 8A on the previous page.**

In the following setup for a mixing problem:



find a formula for the concentration of salt in Tank A after t minutes, in g/L.

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9 (5pts each = 25 points in all) For each of the following, fill in the bubble to decide if the statement is true or false. Correct answers will receive 5 points; answers left blank will receive 2.5 points; and incorrect answers will receive 0 points.

Explanations are not needed and will not be looked at, but writing them out may still help you in your own studying or deciding the correct answer.

(a) Eigenvectors for distinct eigenvalues of a real symmetric matrix are orthogonal.

☐ TRUE ☐ FALSE

(b) If A and B are similar matrices, then A and B have the same characteristic polynomial.

☐ TRUE ☐ FALSE

(c) If A is diagonalizable, then A^T is also diagonalizable, and A^T has the same eigenvalues with the same multiplicity as A .

☐ TRUE ☐ FALSE

(d) For any linear transformation $T: V \rightarrow W$ and vector spaces V and W , If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent, then $\{T(\mathbf{v}_1), T(\mathbf{v}_2), T(\mathbf{v}_3)\}$ is linearly dependent.

☐ TRUE ☐ FALSE

(e) For any linear transformation $T: V \rightarrow W$ and vector spaces V and W , If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent, then $\{T(\mathbf{v}_1), T(\mathbf{v}_2), T(\mathbf{v}_3)\}$ is linearly independent.

☐ TRUE ☐ FALSE

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10 (5pts each = 25 points in all) For each of the following, fill in the bubble to decide if the statement is true or false. Correct answers will receive 5 points; answers left blank will receive 2.5 points; and incorrect answers will receive 0 points.

Explanations are not needed and will not be looked at, but writing them out may still help you in your own studying or deciding the correct answer.

(a) For any $n \times n$ matrix A , the formula $\langle \mathbf{x}, \mathbf{y} \rangle = (A\mathbf{x}) \cdot (A\mathbf{y})$ is an inner product on $\mathbb{R}^n$.

☐ TRUE ☐ FALSE

(b) If A is an orthogonal matrix, then the linear map $T(\mathbf{x}) = A\mathbf{x}$ is one-to-one and onto.

☐ TRUE ☐ FALSE

(c) All eigenvalues of any non-singular matrix are real numbers.

☐ TRUE ☐ FALSE

(d) For any 5×6 matrix A , if $\text{Col } A \neq \mathbb{R}^5$, then $\text{Nul } A$ is at least 2-dimensional

☐ TRUE ☐ FALSE

(e) There is only one 4×3 matrix with rank 3 that is also in reduced echelon form.

☐ TRUE ☐ FALSE

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11 (5pts each = 25 points in all) For each of the following, fill in the bubble to decide if the statement is true or false. Correct answers will receive 5 points; answers left blank will receive 2.5 points; and incorrect answers will receive 0 points.

Explanations are not needed and will not be looked at, but writing them out may still help you in your own studying or deciding the correct answer.

(a) For any $n \times n$ non-diagonalizable real matrix A , if λ is a real eigenvalue of A with eigenvector $\mathbf{u}$, then $\mathbf{x}(t) = e^{\lambda t} \mathbf{u}$ is a solution to $\mathbf{x}'(t) = A\mathbf{x}(t)$.
☐ TRUE ☐ FALSE

(b) For any $n \times n$ real matrix A , if every eigenvalue of A is a complex (non-real) number, then $\mathbf{x}'(t) = A\mathbf{x}(t)$ has no real solutions besides $\mathbf{x}(t) = \mathbf{0}$.
☐ TRUE ☐ FALSE

(c) The dimensions of the row space and of the column space of any matrix A are always the same, whether or not A is square.
☐ TRUE ☐ FALSE

(d) If $f(x)$ is any twice-differentiable odd function, then for any $L > 0$, the Fourier sine series of $f(x)$ on $[0, L]$ is the Fourier series of $f(x)$ on $[-L, L]$.
☐ TRUE ☐ FALSE

(e) If $f(x)$ is any twice-differentiable odd function, then for any $L > 0$, the Fourier cosine series of $f(x)$ on $[0, L]$ is 0.
☐ TRUE ☐ FALSE

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12 (15 points) Find the general solution to the following system of differential equations:

$$\begin{aligned}u'(t) &= -u(t) - v(t) - v'(t), \\v''(t) &= u(t) - 4v(t) - 2v'(t).\end{aligned}$$

Hint: One of the roots of the cubic is -1 .

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Scratch work